

Field Weakening Guide

Power Application Controller™



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1 DESCRIPTION

For PMSM, field weakening is used to achieve a higher motor speed after the inverter reaches the maximum voltage output. The principle of the field weakening is to inject a negative d-axis current to reduce the motor flux. Therefore, a higher motor speed is achievable with the same amount of voltage provided by the inverter. It is worthy to point out that the motor torque could be reduced because of the weakened motor flux, hence it is recommended to enable the field weakening only after the inverter reaches the PWM saturation (maximize the voltage output).

With Qorvo FOC firmware, the field weakening can be enabled for the speed mode only. For the torque mode, the field weakening is achieved by commanding the negative I_{dref} CL (A), which is a customer input parameter. The end user has the full control on this command to meet the application needs.

2 FIRMWARE FLOW

The field weakening controller is illustrated in Figure 1. The inputs to the field weakening block include the speed reference, speed feedback and PWM saturation signal. The output will provide the Idref for the inner current controller.

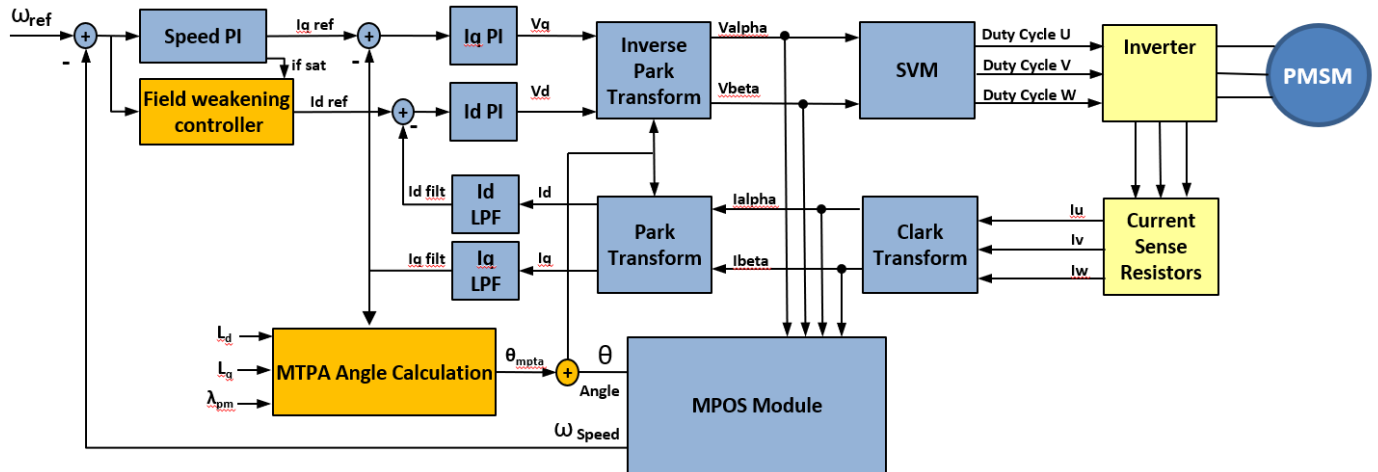


Figure 1: FOC control diagram with Field Weakening Controller

3 USING FIELD WEAKENING FEATURE

3.1 Enabling Field Weakening

To enable the field weakening feature, find the compile option in config_features.h and verify that it is defined:

```
#define ENABLE_FIELD_WEAKENING
```

3.2 Firmware Options

To configure the field weakening, please verify that the desired options are defined in config_features.h and that the selections and values are appropriate for the application:

Field Weakening Configuration

There are two field weakening control methods: Incremental control and PID control. The incremental control applies the hysteresis control to generate the Idref with the input of the speed error. For the PID control, a PI controller is used to get the desired Idref. Hence, more tuning effort is required with PID method than Incremental method.

Incremental control

OUTERLOOP_ERR_THRESHOLD

This applies a threshold for the hysteresis controller.

ID_INCREMENT_BIG

This generates a large incremental step on Idref when the speed error between the reference and feedback is big.

ID_INCREMENT_SMALL

This generates a small incremental step on Idref when the speed error between the reference and feedback is small. This essentially achieves a fine-tuning purpose.

PID control

VRMS_FILTER_CUTOFF_FREQ_Q16

This applies a cut-off frequency for the inverter voltage feedback.

3.3 Field Weakening internal variables Display

The field weakening variables can be exported in Parameter group 25 and 26 as below for display purposes. The debug feature needs to be enabled in config_debug.h:

```
// Field Weakening
#ifdef ENABLE_FIELD_WEAKENING
    case 25:
        parameter_value1 = motor_ptr->fw.id_sp;
        parameter_value2 = fix16_mul(motor_ptr->fw.spd_threshold,
FIX16_1P0_DIV_2_PI);
        break;
    case 26:
        #if (FIELD_WEAKENING_METHOD == PID_CONTROL)
            parameter_value1 = motor_ptr->fw.fw_pid.ka_q16;
            parameter_value2 = motor_ptr->fw.fw_pid.kb_q16;
            parameter_value3 = motor_ptr->fw.vmax_scl_q14 << 2;
        #endif
        break;
    case 27:
        break;
#endif
```

3.4 Tuning Parameters (GUI)

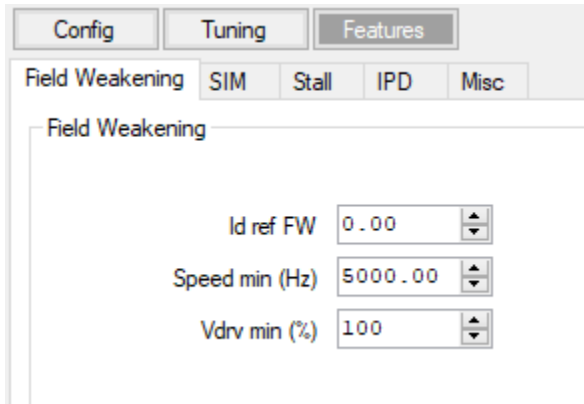
GUI configurable parameters are available as below for the Speed mode only:

Incremental control

Id ref FW: Set the max d-axis current (Negative) reference. The more negative current is, the stronger the field weakening would be.

Speed min(Hz): Configure the min speed to enable the field weakening controller. If the speed is less than this setting, the field weakening is not enabled.

Vdrv min(%): This parameter is obsolete in the firmware.



Field Weakening	
Id ref FW	0.00
Speed min (Hz)	5000.00
Vdrv min (%)	100

PID control

Id ref FW: Set the max d-axis current (Negative) reference. The more negative current is, the stronger the field weakening would be.

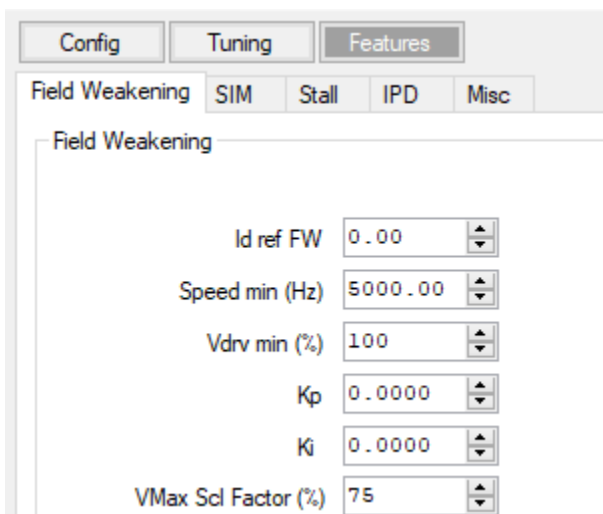
Speed min(Hz): Configure the min speed to enable the field weakening controller. If the speed is less than this setting, the field weakening is not enabled.

Vdrv min(%): This parameter is obsolete in the firmware.

Kp: The proportional gain for the PI controller.

Ki: The integral gain for the PI controller.

Vmax Scl Factor(%): The voltage margin (Per Unit) of the inverter max output. It is typically between 75% and 100%.



Field Weakening	
Id ref FW	0.00
Speed min (Hz)	5000.00
Vdrv min (%)	100
Kp	0.0000
Ki	0.0000
VMax Scl Factor (%)	75



CONTACT INFORMATION

For the latest specifications, additional product information, worldwide sales and distribution locations:

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